

Chapter 6

Exponential Functions

An exponential function puts the variable in the exponent,

$$f(x) = ab^x,$$

with a fixed base $b > 0$ and a starting value a . Each step of one unit in x multiplies the output by the same factor b , in contrast to a line, where each step adds the same amount. Repeated multiplication is the engine of growth when $b > 1$ and of decay when $0 < b < 1$, and it governs populations, cooling, interest, and the decay of radioactive matter.

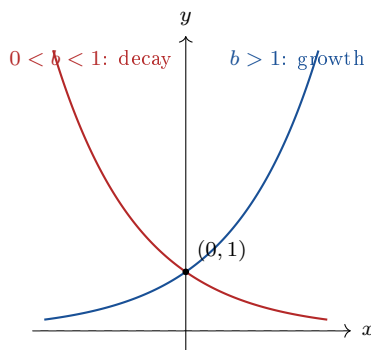
6.1 Constant Ratio, Not Constant Difference

A linear function changes by a constant difference: each unit step adds m . An exponential function changes by a constant ratio: each unit step multiplies by b . This single distinction separates the two kinds of growth and decides which model a data cloud calls for.

Definition: Exponential function. An *exponential function* has the form $f(x) = ab^x$ with base $b > 0$, $b \neq 1$, and $a \neq 0$. The value $a = f(0)$ is the output at $x = 0$; the base b is the factor by which the output multiplies for each unit increase in x .

Key idea. Linear growth adds a fixed amount per step; exponential growth multiplies by a fixed factor per step. Over equal input steps, a linear function has equal differences and an exponential function has equal ratios.

Example 6.1.1. (*spotting the pattern in a table*) The outputs 3, 6, 12, 24, 48 at $x = 0, 1, 2, 3, 4$ multiply by 2 each step, so they fit $f(x) = 3 \cdot 2^x$. The outputs 3, 5, 7, 9, 11 instead add 2 each step, a linear pattern $f(x) = 2x + 3$. Reading whether successive outputs share a difference or a ratio identifies the model.



Every exponential b^x passes through $(0, 1)$, since $b^0 = 1$, and stays strictly positive, since a positive base raised to any power is positive. The x -axis is a horizontal asymptote: the graph approaches it on one side without reaching it.

6.2 Negative Exponents and Decay

A negative exponent is a reciprocal: $b^{-x} = \frac{1}{b^x} = \left(\frac{1}{b}\right)^x$. So raising a base to negative powers is the same as raising its reciprocal to positive powers, which turns growth into decay. A base larger than 1 run with negative exponents shrinks; equivalently, a base between 0 and 1 shrinks with positive exponents. The two descriptions are the same function.

Key idea. $b^{-x} = \left(\frac{1}{b}\right)^x$. A negative exponent reflects the graph across the y -axis and converts growth into decay. A decay function can be written with a base above 1 and a negative exponent, or with a base below 1 and a positive exponent; both name the same curve.

Example 6.2.1. (*two ways to write decay*) The function $f(x) = 2^{-x}$ halves at each step: $1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \dots$. The same outputs come from $g(x) = \left(\frac{1}{2}\right)^x$. Writing the decay with base $\frac{1}{2}$ and positive exponent, or base 2 and negative exponent, is a matter of preference; the graph is identical.

6.3 The Number e

One base stands out, and it arises from asking how fast money grows under continual compounding. Invest one dollar for one year at 100% annual interest. Compounded once, the year-end value is $(1 + 1) = 2$ dollars. Compounded twice, each half-year adds 50%: $(1 + \frac{1}{2})^2 = 2.25$. Compounded n times, the year-end factor is

$$\left(1 + \frac{1}{n}\right)^n.$$

As the compounding is split into ever finer periods, this factor does not run away to infinity, nor does it stall; it climbs and settles toward a fixed ceiling.

compounding n	year-end factor $(1 + \frac{1}{n})^n$
1 (annually)	2.000000
2 (semiannually)	2.250000
4 (quarterly)	2.441406
12 (monthly)	2.613035
52 (weekly)	2.692597
365 (daily)	2.714567
8760 (hourly)	2.718127
1,000,000	2.718280

The values press upward against a ceiling near 2.71828, never crossing it however fine the compounding. That ceiling is the number e , an irrational constant

$$e = 2.718281828\dots,$$

the value the year-end factor approaches under continuous compounding.

Key idea. The number e is the ceiling the compounding factor $(1 + \frac{1}{n})^n$ rises toward as the compounding is made continuous. It is the natural base for exponential growth and decay, written $f(x) = a e^{kx}$, where $k > 0$ gives growth and $k < 0$ gives decay.

The picture. Splitting the year into finer compounding periods adds a little more growth each time, but each addition is smaller than the last, and the running total is trapped beneath a fixed height. That height is e . The continuous-growth curve e^x is the smooth result of compounding at every instant.

Example 6.3.1. (*continuous growth*) A culture growing continuously at rate $k = 0.5$ per hour from an initial 200 cells is modeled by $P(t) = 200 e^{0.5t}$. After 3 hours, $P(3) = 200 e^{1.5} \approx 200(4.4817) \approx 896$ cells. The base e encodes growth happening at every instant rather than in discrete steps.

6.4 Change of Base

Any exponential can be rewritten in any other base, because every positive base is a power of every other. In particular, every exponential can be written with base e , which is why e suffices as a universal base. Writing $b = e^k$ for the exponent k that produces b , the conversion follows from the exponent rules.

Key idea. Change of base into e . For a positive base b , let k be the exponent with $e^k = b$. Then

$$b^x = (e^k)^x = e^{kx},$$

so every exponential is a base- e exponential with a rescaled exponent. The value k is denoted $\ln b$, the natural logarithm of b , studied next; here it is simply the exponent that turns e into b .

Example 6.4.1. (*rewriting in base e*) Write $f(x) = 3^x$ in base e . The exponent producing 3 is $k = \ln 3 \approx 1.0986$ (so $e^{1.0986} \approx 3$), giving

$$3^x = (e^{\ln 3})^x = e^{(\ln 3)x} \approx e^{1.0986x}.$$

The growth base 3 becomes the continuous rate $k \approx 1.0986$.

Example 6.4.2. (*a decay rate in base e*) A substance retaining 90% of its mass per year obeys $f(t) = (0.9)^t$. The exponent producing 0.9 is $k = \ln 0.9 \approx -0.1054$, so $f(t) \approx e^{-0.1054t}$. The negative continuous rate confirms decay.

6.5 Half-Life and Radioactive Decay

A radioactive element loses a fixed *fraction* of its mass in each fixed span of time, which is exactly constant-ratio behavior, so its mass is an exponential function of time. The natural span to measure is the *half-life*: the time for the mass to fall to half its value. After one half-life, half remains; after two, a quarter; after three, an eighth.

Definition: Half-life. The *half-life* T of a decaying quantity is the time required for it to diminish to one-half of its current amount. Starting from M_0 , the amount after time t is

$$M(t) = M_0 \left(\frac{1}{2}\right)^{t/T},$$

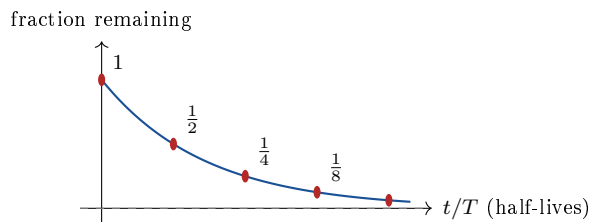
since t/T counts how many half-lives have elapsed.

Key idea. Half-life decay is constant-ratio decay with ratio $\frac{1}{2}$ per half-life. The exponent t/T is the number of half-lives elapsed; each whole number of half-lives multiplies the remaining amount by another factor of $\frac{1}{2}$.

Example 6.5.1. (*carbon-14 dating*) Carbon-14 has a half-life of about 5730 years. A sample starting with M_0 grams has

$$M(t) = M_0 \left(\frac{1}{2}\right)^{t/5730}.$$

After 11,460 years, two half-lives have passed, so $M = M_0 \left(\frac{1}{2}\right)^2 = \frac{1}{4}M_0$. If a sample is measured at 25% of its original carbon-14, it has aged two half-lives, about 11,460 years.



Example 6.5.2. (*solving for elapsed time without logarithms*) A medical isotope has a half-life of 6 hours. A dose of 80 milligrams is given; how much remains after 18 hours? Eighteen hours is $18/6 = 3$ half-lives, so

$$M = 80 \left(\frac{1}{2}\right)^3 = 80 \cdot \frac{1}{8} = 10 \text{ mg.}$$

When the elapsed time is a whole number of half-lives, the answer needs no logarithms: just halve repeatedly.

Common pitfall. Half-life decay multiplies by $\frac{1}{2}$ each half-life; it does not subtract a fixed amount. After one half-life 80 mg becomes 40, after another it becomes 20, not 0. Treating decay as repeated subtraction, rather than repeated halving, is a common error and predicts the substance vanishing far too soon.

Interactive. In the Function Fit Lab, set the parent function to an exponential and tune $A f(b(x - h)) + k$ to thread a noisy point cloud. A positive horizontal scale grows and a negative one decays; matching the curve to data is the practical form of choosing a growth or decay rate. Compare how altering the base steepens or flattens the climb.

https://bobovski66.github.io/Precalculus/demos/function_fit.html

6.6 Exercises

Exercises 6.6

- 6.6.1. Decide whether each table is linear or exponential, and give a formula: (a) outputs 5, 10, 20, 40 at $x = 0, 1, 2, 3$; (b) outputs 5, 8, 11, 14 at the same inputs; (c) outputs 100, 20, 4 at $x = 0, 1, 2$.
- 6.6.2. Sketch $f(x) = 3^x$ and $g(x) = 3^{-x}$ on the same axes, mark the shared point $(0, 1)$, and identify the horizontal asymptote of each.
- 6.6.3. Rewrite each as a positive-base, positive-exponent decay or growth, whichever applies: (a) 5^{-x} ; (b) $(\frac{1}{4})^x$; (c) 2^{-3x} .
- 6.6.4. Using the compounding table, explain in your own words why $(1 + \frac{1}{n})^n$ does not grow without bound as the compounding is refined, and state the value it approaches.
- 6.6.5. A bank account of \$1000 grows continuously at 4% per year, modeled by $A(t) = 1000e^{0.04t}$. Find the balance after 10 years.
- 6.6.6. Rewrite $f(x) = 4^x$ in base e , giving the continuous rate $k = \ln 4$ to four decimals.
- 6.6.7. Rewrite $f(t) = (0.8)^t$ in base e and state whether the continuous rate is positive or negative, with its meaning.
- 6.6.8. A radioactive sample has a half-life of 10 years. Starting from 64 grams, find the amount remaining after 10, 20, and 30 years without using logarithms.
- 6.6.9. Write the decay model $M(t) = M_0 (\frac{1}{2})^{t/T}$ for an element with half-life $T = 1600$ years, and find the fraction remaining after 4800 years.
- 6.6.10. A sample is measured at one-eighth of its original radioactive content. Determine how many half-lives have elapsed, and if the half-life is 30 years, the sample's age.
- 6.6.11. Explain why a radioactive element's mass is an exponential function of time, referring to the constant-ratio property and the meaning of half-life.
- 6.6.12. Two isotopes start with equal mass; one has half-life 2 years, the other 4 years. Compute the fraction of each remaining after 8 years, and explain which

has decayed further and why.